Claims

1. A method for assessing a subject with suspected infection comprising the steps of:
   (a) determining the amount of a first biomarker in a sample of the subject, said first biomarker being PCT;
   (b) determining the amount of a second biomarker in a sample of the subject, wherein said second biomarker is selected from the group consisting of: a cardiac Troponin, Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase
   (c) comparing the amounts of the biomarkers to references for said biomarkers and/or calculating a score for assessing the subject with suspected infection based on the amounts of the biomarkers; and
   (d) assessing said subject based on the comparison and/or the calculation made in step (c).

2. The method of claim 1, wherein in step (b)
   (i) if the amount of a cardiac Troponin is determined as the second biomarker, the method will further comprise determining the amount of Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine as a third biomarker; or
   (ii) if the amount of Creatinine is determined as the second biomarker, the method will further comprise determining the amount of Aspartate aminotransferase or Bilirubin as a third biomarker, or
   (iii) if the amount of sTREM1 is determined as the second biomarker, the method will further comprise determining the amount of Aspartate aminotransferase or HBP as a third biomarker.

3. The method of claim 1 or 2, wherein the subject is a subject presenting at the emergency department.

4. The method of any one of claims 1 to 3, wherein the assessment is the assessment of the risk of developing sepsis and/or the assessment of the risk that the subject's condition of the subject will deteriorate.

5. The method of any one of claims 1 to 4, wherein said references are references for each biomarker derived from at least one subject known to be at risk for developing sepsis, preferably wherein amounts for each of the biomarkers being essentially identical or

similar to the corresponding references are indicative for a subject being at risk for developing sepsis while amounts for each of the biomarkers being different from the corresponding references are indicative for a subject being not at risk for developing sepsis.

6. The method of any one of claims 1 to 4, wherein said references are references for each biomarker derived from at least one subject known not to be at risk for developing sepsis, preferably wherein amounts for each of the biomarkers being essentially identical or similar to the corresponding references are indicative for a subject being not at risk for developing sepsis while amounts for each of the biomarkers being different from the corresponding references are indicative for a subject being at risk for developing sepsis.

7. The method of any one of claims 1 to 6, wherein said subject suffers from an infection or is suspected to suffer from an infection and/or wherein the subject is a human.

8. The method of any one of claims 1 to 7, wherein said sample is a blood sample or a sample derived therefrom.

9. A computer-implemented method for assessing a subject with suspected infection comprising the steps of:
   (a) receiving a value for the amount of a first biomarker in a sample of the subject, said first biomarker being PCT;
   (b) receiving a value for the amount of a second biomarker in a sample of the subject, wherein said second biomarker is selected from the group consisting of: a cardiac Troponin, Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase;
   (c) comparing the values for the amounts of the biomarkers to references for said biomarkers and/or calculating a score for assessing the subject with suspected infection based on the amounts of the biomarkers; and
   (d) assessing said subject based on the comparison and/or the calculation made in step (c).

10. The method of claim 9, wherein in step (b)
   (i) if the value for the amount of a cardiac Troponin is received as the second biomarker, the method will further comprise receiving a value for the amount of Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine as a third biomarker; or

(ii) if the value for the amount of Creatinine is received as the second biomarker, the method will further comprise receiving a value for the amount of Aspartate aminotransferase or Bilirubin as a third biomarker, or

(iii) if the value for the amount of sTREM1 is received as the second biomarker, the method will further comprise receiving a value for the amount of Aspartate aminotransferase or HBP as a third biomarker.

11. A device for assessing a subject with suspected infection comprising:

(a) a measuring unit for determining the amount of a first biomarker being PCT and a second biomarker selected from the group consisting of: a cardiac Troponin, Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase in a sample of the subject, said measuring unit comprising a detection system for the first biomarker and the second biomarker; and

(b) an evaluation unit operably linked to the measuring unit comprising a database with stored references for the first biomarker and the second biomarker, preferably, as specified in any one of claims 1 to 8 and a data processor comprising instructions for carrying out a comparison of the amount of the first biomarker and the second biomarker to references and/or for carrying out a calculation of a score for assessing the subject with suspected infection based on the amounts of the biomarkers, preferably, as specified in any one of claims 1 to 9 and for assessing said subject based on the comparison, said evaluation unit being capable of automatically receiving values for the amounts of the biomarkers from the measuring unit,

wherein optionally said measuring unit determines and comprises a detection system for a third biomarker and wherein said database comprises stored a reference for a third biomarker, said third biomarker being

(i) if a cardiac Troponin is the second biomarker, Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine; or

(ii) if Creatinine is the second biomarker, Aspartate aminotransferase or Bilirubin, or

(iii) if sTREM1 is the second biomarker, Aspartate aminotransferase or HBP, and

wherein optionally said detection system comprises at least one detection agent being capable of specifically detecting each of the biomarkers.

12. A device for assessing a subject with suspected infection comprising an evaluation unit comprising a database with stored references for a first biomarker being PCT and a second biomarker is selected from the group consisting of: a cardiac Troponin, Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase and a data processor comprising instructions for carrying out a comparison of the amount of the first biomarker and the second biomarker to references, preferably, as specified in any one of claims 1 to 11 and for assessing said subject based on the comparison, said evaluation unit being capable of receiving values for the amounts of the biomarkers determined in a sample of the subject,

wherein optionally said database comprises a stored reference for a third biomarker, said third biomarker being

(i) if a cardiac Troponin is the second biomarker, Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine; or

(ii) if Creatinine is the second biomarker, Aspartate aminotransferase or Bilirubin, or

(iii) if sTREM1 is the second biomarker, Aspartate aminotransferase or HBP.

13. Use of i) a first biomarker being PCT and a second biomarker selected from the group consisting of a cardiac Troponin, Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase, or i) an detection agent specifically binding to said first biomarker and an detection agent specifically binding to said second biomarker for assessing a subject with suspected infection,

wherein optionally a third biomarker or an detection agent specifically binding to said third biomarker is used in addition, said third biomarker being

(i) if a cardiac Troponin is the second biomarker Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine; or

(ii) if Creatinine is the second biomarker, Aspartate aminotransferase, or

(iii) if sTREM1 is the second biomarker, Aspartate aminotransferase or HBP.

14. A kit for assessing a subject with suspected infection comprising an detection agent specifically binding to a first biomarker being PCT and an detection agent specifically binding to a second biomarker selected from the group consisting of a cardiac Troponin,

Creatinine, a BNP-type peptide, sTREM1, ESM-1, Haptoglobin, Heparin binding protein (HBP) and Aspartate aminotransferase,
wherein optionally said kit further comprises an detection agent specifically binding a third biomarker, said third biomarker being

(i) if a cardiac Troponin is the second biomarker, Bilirubin, Aspartate aminotransferase, Alanine aminotransferase, Haptoglobin, ESM1 or Creatinine; or

(ii) if Creatinine is the second biomarker, Aspartate aminotransferase or Bilirubin, or

(iii) if sTREM1 is the second biomarker, Aspartate aminotransferase or HBP.

15. The method, device, use or kit of any one of the preceding claims, wherein the assessment is the assessment of the risk of developing sepsis, and wherein optionally the risk of developing sepsis within 48 hours is predicted.